

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12414716
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Muehlsteff; Jens et al.

Photoplethysmography pulse oximeter for cardiopulmonary resuscitation

Abstract

A PPG pulse oximeter employing a dual PPG probe including a central PPG sensor and a peripheral PPG sensor connectable in a circuit configuration with a pulse oximeter monitor. In operation, the pulse oximeter monitor controls synchronous generations of a central PPG signal by the central PPG sensor and of a peripheral PPG signal by the peripheral PPG sensor, and also control a CPR pulse detection via the dual PPG probe including a detection of a presence of a spontaneous pulse of the central PPG signal and a detection of a presence of a spontaneous pulse of the peripheral PPG signal.

Inventors:	Muehlsteff; Jens (Aachen, DE), Wijshoff; Ralph Wilhelm Christianus Gemma Rosa (Eindhoven, NL)
Applicant:	KONINKLIJKE PHILIPS N.V. (Eindhoven, NL)
Family ID:	1000008820377
Assignee:	Koninklijke Philips N.V. (Eindhoven, NL)
Appl. No.:	17/260059
Filed (or PCT Filed):	July 12, 2019
PCT No.:	PCT/EP2019/068831
PCT Pub. No.:	WO2020/011982
PCT Pub. Date:	January 16, 2020

Prior Publication Data

Document Identifier	Publication Date
US 20210282670 A1	Sep. 16, 2021

Related U.S. Application Data

us-provisional-application US 62697561 20180713

Publication Classification

Int. Cl.: A61B5/1455 (20060101); A61B5/00 (20060101); A61B5/024 (20060101)

U.S. Cl.:

CPC A61B5/1455 (20130101); A61B5/02416 (20130101); A61B5/742 (20130101); A61B2505/01 (20130101)

Field of Classification Search

CPC: A61B (5/1455); A61B (5/02416); A61B (5/742); A61B (2505/01)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2004/0039269	12/2003	Ward	600/312	A61B 5/0086
2004/0267324	12/2003	Boucher et al.	N/A	N/A
2010/0081892	12/2009	Addison et al.	N/A	N/A
2013/0324808	12/2012	Abdul-Hafiz et al.	N/A	N/A
2015/0099951	12/2014	Dejong et al.	N/A	N/A
2015/0380875	12/2014	Coverston et al.	N/A	N/A
2016/0206504	12/2015	Giarracco	N/A	A61B 5/14551
2016/0374623	12/2015	Muehlsteff et al.	N/A	N/A
2017/0027488	12/2016	Enenkel	N/A	A61B 5/443

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2011086520	12/2010	WO	N/A

OTHER PUBLICATIONS

Spittal, M.J.. "Evaluation of pulse oximetry during cardiopulmonary resuscitation". Anaesthesia, vol. 48, No. 8, (Aug. 1, 1993), pp. 701-703. cited by applicant

International Search Report for PCT/EP2019/068831 dated Jul. 12, 2019. cited by applicant

Berg, R.A. et al., "Adverse Hemodynamic Effects of Interrupting Chest Compressions for Rescue Breathing During Cardiopulmonary Resuscitation for Ventricular Fibrillation Cardiac Arrest". Circulation, Journal of the American Heart Association. 201;104:2465-2470. cited by applicant

Christenson, J. et al., "Chest Compression Fraction Determines Survival in Patients with Out-of-Hospital Ventricular Fibrillation". Circulation, Journal of the American Heart Association. 2009;120:1241-1247. cited by applicant

Link, M.S. et al., "Part 7: Adult Advanced Cardiovascular Life Support—2015 American Heart Association Guidelines Update for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care". Circulation, Journal of the American Heart Association. 2015;132[suppl 2]:S444-S464. cited by applicant

Cummingham, L.M. et al., “Cardiopulmonary resuscitation for cardiac arrest: the importance of uninterrupted chest compressions in cardiac arrest resuscitation”. American Journal of Emergency Medicine (2012) 30, 1630-1638. cited by applicant

Eberle, B. et al., “Checking the carotid pulse check: diagnostic accuracy of first responders in patients with and without a pulse”. Resuscitation 33 (1996) 107-116. cited by applicant

Hubner, P. et al., “Pulse appearance in photoplethysmography signals obtained from finger, nose and ear during extracorporeal life support”. Abstracts/Resuscitation 96S(2015) 43-157. cited by applicant

Muehlsteff, J. et al., “Continuous Cuff-less Blood Pressure Monitoring based on the Pulse Arrival Time Approach: The Impact of Posture”. Philips Research Europe, 2008. cited by applicant

Moule, P. “Checking the carotid pulse: diagnostic accuracy in students of the healthcare professions”. Resuscitation 44 (2000) 195-201. cited by applicant

Ochoa, F. Javier et al., “Competence of health professionals to check the carotid pulse”. Resuscitation 37 (1998) 173-175. cited by applicant

Payne, R.A. et al., “Pulse transit time measured from the ECG: an unreliable marker of beat-to-beat blood pressure”. J Appl Physiol 100: 136-141, 2006. cited by applicant

Wijshoff, R. et al., “Detection of a spontaneous pulse in photoplethysmograms during automated cardiopulmonary resuscitation in a porcine model”. Resuscitation 84 (2013) 1625-1632. cited by applicant

Soar, J. et al., “European Resuscitation Council Guidelines for Resuscitation 2015 Section 3. Adult advanced life support”. Resuscitation 95 (2015) 100-147. cited by applicant

Tibballs, J. et al., “Reliability of pulse palpation by healthcare personnel to diagnose paediatric cardiac arrest”. Resuscitation 80 (2009) 61-64. cited by applicant

Wijshoff, R. et al., “Photoplethysmography-Based Algorithm for Detection of Cardiogenic Output During Cardiopulmonary Resuscitation”. IEEE Transaction on Biomedical Engineering, vol. 62, No. 3, Mar. 2015. cited by applicant

Wik, L. et al., “Quality of Cardiopulmonary Resuscitation During Out-of-Hospital Cardiac Arrest”. JAMA, Jan. 19, 2005—vol. 293, No. 3, pp. 299-304. cited by applicant

Primary Examiner: Valvis; Alex M

Assistant Examiner: Haney; Jonathan M

Background/Summary

(1) This application is a national stage application under 35 U.S.C. § 371 of International Application No. PCT/EP2019/068831 filed on Jul. 12, 2019 and published in the English language on Jan. 16, 2020 as International Publication No. WO2020/011982, which claims priority to U.S. Patent Application No. 62/697,561 filed on Jul. 13, 2018, the entire disclosures of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) The present disclosure relates generally to photoplethysmography for detecting volumetric changes in a peripheral circulation of blood. The present disclosure specifically relates to an application of photoplethysmography to cardiopulmonary resuscitation.

BACKGROUND OF THE INVENTION

(2) Cardiopulmonary resuscitation (CPR) is an emergency procedure for a person suffering from a cardiac arrest whereby a goal of CPR is to achieve return of spontaneous circulation (ROSC). During CPR, chest compressions are delivered to artificially generate circulation of blood, and

ventilations are given to supply blood with oxygen. When ROSC has been achieved, the heart of the cardiac arrested person has resumed beating and generates a spontaneous circulation which is life-sustaining. CPR may therefore be stopped after achieving ROSC.

(3) Determining whether the cardiac arrested person has achieved ROSC involves checking whether the person has a spontaneous circulatory pulse, i.e., whether the heart is beating and generating output. Currently in the art of the present disclosure, pulse checks are typically performed by manual palpation. However, manual palpation is known to be unreliable and time-consuming, and requires interruption of the chest compressions. Manual palpation therefore may take significantly longer than the ten (10) seconds recommended by the CPR guidelines for ROSC assessment, especially when a spontaneous pulse is absent. As a result, manual palpation may lead to long interruptions in the chest compressions, which reduces the compression-generated blood flow and may thereby negatively impact CPR outcome.

(4) Pulse detection may be supported by using photoplethysmography (PPG), which is a readily available, non-invasive and easy-to-use optical technology currently widely applied in pulse oximetry. More particularly, pre-clinical results in the art of the present disclosure support the feasibility of PPG-based pulse detection during CPR.

SUMMARY OF THE INVENTION

(5) In accordance with exemplary embodiments of the present disclosure, the inventions of the present disclosure are derived from a recognition that, when a heart of a cardiac arrested person resumes beating, a spontaneous pulse of the person may be observed directly in a PPG signal measured at a central location of the person (e.g., nose, ear or forehead), and the spontaneous pulse of the person may be subsequently observed in a PPG signal measured at a peripheral location of the person (e.g., finger or toe) with a delay of about sixty (60) seconds. As such, the inventions of the present disclosure premised in view of a central PPG signal being better suited to an early detection of a presence of a spontaneous pulse of the person and further in view of a peripheral PPG signal providing a qualitative measure of the level of circulation of the person (i.e., when a spontaneous pulse of the person is observed peripherally, the inventions of the present disclosure indicate that some minimum level of circulation has been achieved by that person).

(6) More particularly, the inventions of the present disclosure provide devices, controls and methods incorporating two (2) PPG sensors to support pulse detection during CPR. One PPG sensor is placed centrally on a cardiac arrested person to obtain an early indication of presence of a spontaneous pulse, and the other PPG sensor is placed peripherally on the person to obtain a qualitative measure of the level of circulation of the person. Consequently, the central PPG measurement may advantageously provide clinical decision support regarding administration of medication during the CPR, and the peripheral PPG measurement may advantageously provide clinical decision support regarding when to stop chest compressions and further assess for a potential ROSC.

(7) One embodiment of the inventions of the present disclosure is a PPG pulse oximeter comprising a dual PPG probe including a central PPG sensor and a peripheral PPG sensor. The PPG pulse oximeter further comprises a pulse oximeter monitor connectable to the central PPG sensor and the peripheral PPG sensor in a circuit configuration to (1) control synchronous generations of a central PPG signal by the central PPG sensor and of a peripheral PPG signal by the peripheral PPG sensor and (2) also control a CPR pulse detection by the dual PPG probe including a detection of a presence of a spontaneous pulse of the central PPG signal and a detection of a presence of a spontaneous pulse of the peripheral PPG signal.

(8) A second embodiment of the inventions of the present disclosure is a PPG pulse monitoring method. The method involves (1) identifying, by a pulse oximeter monitor, a connection of the pulse oximeter monitor to a dual PPG probe in a circuit configuration including a central PPG sensor and a peripheral PPG sensor, (2) controlling, by the pulse oximeter monitor, synchronous generations of a central PPG signal by the central PPG sensor and of the peripheral PPG signal by

the peripheral PPG sensor, and (3) controlling, by the pulse oximeter monitor, a CPR pulse detection by the dual PPG probe including detecting a presence of a spontaneous pulse of the central PPG signal and detecting a presence of a spontaneous pulse of the peripheral PPG signal.

(9) For purposes of describing and claiming the inventions of the present disclosure, the term “pulse oximeter” broadly encompasses all devices, known prior to and subsequent to the present disclosure, for monitoring an oxygen saturation (SO_{sub.2} or SpO_{sub.2}) of a person and optionally monitoring a pulse rate of the person. Examples of a “pulse oximeter” include, but are not limited to Philips FAST SpO₂ as available for the Philips IntelliVue series.

(10) Also for purposes of describing and claiming the inventions of the present disclosure, (1) the term “PPG pulse oximeter” broadly encompasses all pulse oximeters incorporating inventive principles of the present disclosure as exemplary described herein, and the term “PPG pulse monitoring method” broadly encompasses all pulse oximetry based methods that incorporate the inventive principles of the present disclosure as exemplary described herein; (2) terms of the art including, but not limited to, “CPR”, “PPG”, “probe”, “sensor”, “monitor”, “light emitter”, “light emitting diode”, “light detector”, “photodetector”, “optical fiber”, “connector”, “socket” and “microcontroller” are to be interpreted as understood in the art of the present disclosure and as exemplary described herein; (3) the descriptive labeling of the terms “probe”, “sensor”, “light”, “light emitter”, “light emitting diode”, “light detector” and “photodetector” as either “central” or “peripheral” facilitates a distinction between such terms as described and claimed herein without specifying or implying any additional limitation to the terms; (4) the term “application module” broadly encompasses a component of a microcontroller including an electronic circuit and/or an executable program (e.g., executable software and/or firmware stored on non-transitory computer readable medium(s)) for executing a specific application. Any descriptive labeling of an application module herein (e.g., a “pulse width modulator” module, “ROSC detector” module and a “ROSC indicator”) serves to identify a particular application module as described and claimed herein without specifying or implying any additional limitation to the term “application module”; and (5) the term “signal” broadly encompasses all forms of a detectable physical quantity or impulse (e.g., voltage, current, or magnetic field strength) as understood in the art of the present disclosure and as exemplary described herein for communicating information in support of applying various inventive principles of the present disclosure as subsequently described herein. Any descriptive labeling for the term “signal” herein facilitates a distinction between signals as described and claimed herein without specifying or implying any additional limitation to the term “signal”.

(11) The foregoing embodiments and other embodiments of the inventions of the present disclosure as well as various features and advantages of the present disclosure will become further apparent from the following detailed description of various embodiments of the present disclosure read in conjunction with the accompanying drawings. The detailed description and drawings are merely illustrative of the present disclosure rather than limiting, the scope of the present disclosure being defined by the appended claims and equivalents thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIGS. 1A and 1B illustrates an exemplary embodiment of pulse oximeter as known in the art of the present disclosure.
- (2) FIGS. 2A and 2B illustrates exemplary embodiments of a PPG sensor pair in accordance with the inventive principles of the present disclosure.
- (3) FIG. 3 illustrates exemplary PPG signals in accordance with the inventive principles of the present disclosure.
- (4) FIGS. 4A and 4B illustrates a first exemplary embodiment of PPG pulse oximeter in accordance

with the inventive principles of the present disclosure.

(5) FIGS. 5A and 5B illustrates a second exemplary embodiment of PPG pulse oximeter in accordance with the inventive principles of the present disclosure.

(6) FIGS. 6A and 6B illustrates a third exemplary embodiment of PPG pulse oximeter in accordance with the inventive principles of the present disclosure.

(7) FIG. 7 illustrates a flowchart representative of an exemplary embodiment of a PPG pulse monitoring method in accordance with the inventive principles of the present disclosure.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(8) Referring to FIGS. 1A and 1B, a pulse oximeter 20 as known in the art of the present disclosure employs pulse oximeter monitor 30 and a pulse oximeter probe 50 having a connector 60 for plugging into a socket 40 of pulse oximeter monitor 30 to thereby couple a light emitter 51 (e.g., an infra-red light emitting diode) and a light emitter 52 (e.g., a red light-emitting diode) of an SpO2 sensor of pulse oximeter probe 50 in an anti-parallel configuration to a sensor driver 75 of pulse oximeter monitor 30, and for coupling a light detector 53 (e.g., a photodetector) of the SpO2 sensor of pulse oximeter probe 50 to a signal conditioner 77.

(9) In operation, upon a mounting of pulse oximeter probe 50 unto a peripheral location of a person (e.g., a finger 10 as shown), microcontroller 70 executes a pulse width modulator 73 to control a synchronized emission of light by light emitter 51 and light emitter 52 via respective pulses 76a and 76b applied by sensor driver 75 to light emitter 51 and light emitter 52 as known in the art of the disclosure. In practice, pulse 76a and pulse 76b generate optical pulses of different wavelengths, via light emitter 51 and light emitter 52, respectively.

(10) Light emitter 51, light emitter 52 and light detector 53 are optically coupled within pulse oximeter probe 50 whereby light detector 53 will detect light passing through the peripheral location of a person (e.g., finger 10 as shown) to thereby produce a first wavelength detection signal 54 and a second wavelength detection signal 55 as known in art of the present disclosure. Upon being conditioned by signal conditioner 77, microcontroller 70 executes an SpO2 detector 74a to thereby monitor an oxygen saturation of a person as known in the art of the present disclosure, and further executes a pulse rate detector 74b to thereby monitor a pulse rate of the person as known in the art of the present disclosure. Microcontroller 70 employs additional modules (not shown) for displaying and communicating a status of the monitored oxygen saturation of the person and a status of the pulse rate of the person.

(11) To improve upon the prior art of FIGS. 1A and 1B, as will be further described in the present disclosure, embodiments of the inventions of the present disclosure incorporate two (2) single wavelength PPG sensors (e.g., red, near-infrared or green light emitting diodes) using the dual-wavelength pulse oximetry socket 40 (FIG. 1) of known pulse oximeter. This allows for the development of a dedicated CPR dual-PPG-based pulse detection probe which is compatible with the current sockets of pulse oximeters (e.g., socket 40 of pulse oximeter 20).

(12) To facilitate an understanding of the present disclosure, the following description of FIGS. 2 and 3 teaches inventive principles of PPG sensors in accordance with the inventive principles of the present disclosure. From the description of FIG. 1, those having ordinary skill in the art of the present disclosure will appreciate how to apply the inventive principles of the present disclosure for making and using numerous and various embodiments of PPG sensors of the present disclosure.

(13) Referring to FIG. 2A, the inventions of the present disclosure provide for a peripheral PPG sensor 81 employing an optical coupling of a light emitter 82 (e.g., red, near-infrared or green light emitting diode) and a light detector 83 (e.g., a photodiode) whereby light emitted by light emitter 82 through a peripheral location of a person (e.g., finger 10 as shown) is detected by light detector 83. The inventions of the present disclosure further provide for a central PPG sensor 91 employing an optical coupling of a light emitter 92 (e.g., red, near-infrared, or green light emitting diode) and a light detector 93 (e.g., a photodiode) whereby light emitted by light emitter 92 through a central location of a person (e.g., a nose 11 or an ear 12 as shown, or a forehead) is detected by light

detector **93**.

(14) FIG. 2B illustrates an alternate embodiment of a peripheral PPG sensor **81'** and a central PPG sensor **91'** sharing a common light detector **100** (e.g., a photodetector) that is optically coupled via an optical fiber **101** to light emitter **82** and optically coupled via an optical fiber **102** to light emitter **92**.

(15) FIG. 3 illustrates an exemplary peripheral PPG signal **110** obtained by peripheral PPG sensor **81/81'** from finger **10**, and central PPG signals **111** and **112** obtained by central PPG sensor **91/91'** from a nasal septum of nose **11** and an ear pinna of ear **12**. FIG. 3 shows that when the heart of the patient resumes beating, a spontaneous pulse **114** and **115** respectively may be observed directly in the PPG signals **111** and **112** measured centrally at the nose **11** and the ear **12**, but with a delay of about sixty (60) seconds for a spontaneous pulse **113** in the peripheral PPG signal **110** measured peripherally at the finger **10**. The inventions of the present disclosure are premised on central PPG signals **111** and **112** being better suited to early detect presence of a spontaneous pulse during CPR and peripheral PPG signal **110** providing a more qualitative measure of the level of circulation.

(16) To further facilitate an understanding of the present disclosure, the following description of FIGS. 4-6 teaches inventive principles of PPG pulse oximeters in accordance with the inventive principles of the present disclosure. From the description of FIG. 1, those having ordinary skill in the art of the present disclosure will appreciate how to apply the inventive principles of the present disclosure for making and using numerous and various embodiments of PPG pulse oximeters of the present disclosure.

(17) Referring to FIGS. 4A and 4B, a PPG pulse oximeter **21** of the present disclosure employs a pulse oximeter monitor **31** and a dual PPG probe including a peripheral PPG probe **80** and a central PPG probe **90**. Pulse oximeter monitor **31** represents a retrofit or a redesign of pulse oximeter **30** (FIG. 1A) whereby a connector **61** of the dual PPG probe is pin-compatible with socket **40** of pulse oximeter monitor **31** and further whereby a microcontroller **71** includes a central ROSC indicator **78a** and a peripheral ROSC indicator **78b** in addition to or in replacement of SpO₂ detector **74a** (FIG. 1B) and pulse rate detector **74b** (FIG. 1B). More particular to the retrofit or the redesign of pulse oximeter **30**, in practice, the compatibility of connector **61** of dual PPG probe **80/90** with socket **40** of pulse oximeter monitor **30** (FIG. 1) enables the hardware of pulse oximeter monitor **31** to be identical to the hardware of pulse oximeter monitor **30** with the addition of firmware implementing central ROSC indicator **78a** and peripheral ROSC indicator **78b**. Further in practice, for PPG pulse oximeters of the present disclosure incorporating firmware implementing SpO₂ detector **74a**, pulse rate detector **74b**, central ROSC indicator **78a** and peripheral ROSC indicator **78b**, such PPG pulse oximeters will be capable of differentiating a connection of connector **60** of pulse oximeter probe **50** (FIG. 1A) to socket **40** of pulse oximeter monitor **31** and a connection of connector **61** of dual PPG probe **80/90** to socket **40** of pulse oximeter monitor **31** as will be further described in the present disclosure.

(18) Peripheral PPG probe **80** includes peripheral PPG sensor **81** (FIG. 2A) and central PPG probe **90** includes central PPG sensor **91** (FIG. 2A).

(19) Prior to operation, connector **61** is plugged into a socket **40** to thereby couple a light emitter **82** (e.g., a red, near-infrared or green emitting diode) and a light emitter **92** (e.g., a red, near-infrared or green light-emitting diode) in an anti-parallel configuration to sensor driver **75** of pulse oximeter monitor **31**, and for coupling light detector **83** (e.g., a photodetector) and light detector **93** (e.g., a photodetector) in a parallel configuration to signal conditioner **77**.

(20) In operation, upon a mounting of peripheral PPG probe **80** unto a peripheral location of a person (e.g., finger **10** as shown) and a mounting of central PPG probe **90** unto a central location of the person (e.g., nose **11** or ear **12** as shown, or a forehead), microcontroller **71** executes pulse width modulator **73** to control synchronized emission of light by light emitter **82** and light emitter **92** via respective pulses **76c** and **76d** applied by sensor driver **75** to light emitter **82** and light emitter **92** as known in the art of the disclosure. In practice, pulse **76c** and pulse **76d** may generate

optical pulses of the same wavelength, via light emitter **82** and light emitter **92**, respectively. (21) Light emitter **92** and light detector **93** are optically coupled within central PPG probe **90** whereby light detector **93** will detect light passing through the central location of the person (e.g., nose **11** or ear **12** as shown, or a forehead) to thereby produce central PPG signal **94** indicative of a volumetric change in a central circulation of the person. Upon central PPG signal **94** being conditioned by signal conditioner **77**, microcontroller **71** executes a central ROSC indicator **78a** to indicate an early detection of a beating heart, which may support decision making regarding medication.

(22) Similarly, light emitter **82** and light detector **83** are optically coupled within peripheral PPG probe **80** whereby light detector **83** will detect light passing through the peripheral location of the person (e.g., finger **10** as shown) to thereby produce peripheral PPG signal **84** indicative of a volumetric change in a peripheral circulation of the person. Upon peripheral PPG signal **84** being conditioned by signal conditioner **77**, microcontroller **71** executes a peripheral ROSC indicator **78b** to provide a qualitative indication of the level of the spontaneous circulation thus providing decision support in when to stop CPR and further assess a potential return of spontaneous circulation (ROSC).

(23) Microcontroller **71** employs additional modules (not shown) for displaying and communicating a status of any pulse detection during, preceding and/or succeeding CPR of the person.

(24) Referring to FIGS. 5A and 5B, a PPG pulse oximeter **22** of the present disclosure employs pulse oximeter monitor **31** and a dual PPG probe alternatively including a peripheral PPG probe **80'** and a central PPG probe **90'** as shown in FIG. 2B.

(25) Referring to FIGS. 6A and 6B, a PPG pulse oximeter **23** of the present disclosure employs a pulse oximeter monitor **32**, a dual PPG probe of FIGS. 4A and 4B or FIGS. 5A and 5B, and the pulse oximeter probe **50** of FIGS. 1A and 1B. To this end, information is encoded in a connector **62** of the pulse oximeter probe **50** and a connector **63** of the dual PPG probe via an identification means (e.g., via a coding resistor or an IC) whereby a probe identifier **79** of microcontroller **72** may ascertain whether a pulse oximetry probe or the dual PPG probe is attached to the pulse oximetry monitor **32**. As such, a detection of the type of probe will determine which parts of the firmware will be activated. In case of a detection of a connection to dual PPG probe **80/90** or **80'/90'** via connector **63**, central ROSC detector **78a** and peripheral ROSC detector **78b** will be activated among other elements. Conversely, in case of a detection of connection to pulse oximetry probe **50** via connector **62**, SpO2 detector **74a** and pulse rate detector **74b** will be activated among other elements. As a result, pulse oximeter monitor **32** will automatically switch to "CPR mode" upon connection with dual PPG probe **80/90** or **80'/90'** and automatically switch to "SpO2 mode" upon connection with pulse oximetry probe **50**.

(26) FIG. 7 illustrates a flowchart **200** representing a PPG pulse monitoring method of the present disclosure. Referring to FIG. 7, a stage **S202** of flowchart **200** encompasses a synchronized emitter activation during CPR, preceding CPR and/or succeeding CPR.

(27) A stage **S204** and a stage **S206** of flowchart **200** encompasses an analysis of the PPG signals obtained from two (2) sites. Specifically, a central ROSC indicator **78a** and a peripheral ROSC indicator **78b** as previously described herein include algorithms designed to detect a spontaneous pulse component in the respective PPG signal during CPR, preceding CPR and/or succeeding CPR as known in the art of the present disclosure. Stage **S204** and stage **S206** may utilize additional information about the presence and rate of the chest compressions, derived from, e.g., an accelerometer, trans-thoracic impedance signal, a force signal, or a radar signal. The information about compressions can be used to determine how to analyze the PPG signals. First, a spectral analysis of the PPG signals taking into account all compression-related frequency components. Second, a filtering technique to remove or attenuate all compression-related frequency components from the PPG signals. Third, a technique which relies on a PPG signal acquired during a quiet

period in CPR without any chest compression components.

(28) Furthermore, an optional stage **S208** of flowchart **200** may be applied to track blood pressure changes by analyzing time differences between the cardiac-induced pulses in the two synchronously acquired PPG signals, which can be implemented as a trend indicator for the status of the circulation. Stage **S208** may be incorporated in a central ROSC indicator **78a** or a peripheral ROSC indicator **78b**.

(29) A stage **S210** of flowchart **200** provides for a display or communication of any pulse detection and/or circulation trend indication during CPR.

(30) Once initially implemented, stage **S204-S210** are continuously and simultaneously executed until an interruption or a termination of flowchart **200**.

(31) Referring to FIGS. 1-7, those having ordinary skill in the art will appreciate numerous benefits of the inventions of the present disclosure including, but not limited to, an improvement of pulse oximeters for pulse detection during, preceding and/or succeeding CPR.

(32) The present disclosure disclosed herein has been described with reference to the preferred embodiments. Modifications and alterations may occur to others upon reading and understanding the preceding detailed description. It is intended that the invention be construed as including all such modifications and alterations insofar as they come within the scope of the appended claims or the equivalents thereof.

(33) Further, as one having ordinary skill in the art shall appreciate in view of the teachings provided herein, features, elements, components, etc. disclosed and described in the present disclosure/specification and/or depicted in the appended Figures may be implemented in various combinations of hardware and software, and provide functions which may be combined in a single element or multiple elements. For example, the functions of the various features, elements, components, etc. shown/illustrated/depicted in the Figures can be provided through the use of dedicated hardware as well as hardware capable of executing software in association with appropriate software. When provided by a processor, the functions can be provided by a single dedicated processor, by a single shared processor, or by a plurality of individual processors, some of which can be shared and/or multiplexed. Moreover, explicit use of the term “processor” or “controller” should not be construed to refer exclusively to hardware capable of executing software, and can implicitly include, without limitation, digital signal processor (“DSP”) hardware, memory (e.g., read only memory (“ROM”) for storing software, random access memory (“RAM”), non-volatile storage, etc.) and virtually any means and/or machine (including hardware, software, firmware, combinations thereof, etc.) which is capable of (and/or configurable) to perform and/or control a process.

(34) Moreover, all statements herein reciting principles, aspects, and exemplary embodiments of the present disclosure, as well as specific examples thereof, are intended to encompass both structural and functional equivalents thereof. Additionally, it is intended that such equivalents include both currently known equivalents as well as equivalents developed in the future (e.g., any elements developed that can perform the same or substantially similar functionality, regardless of structure). Thus, for example, it will be appreciated by one having ordinary skill in the art in view of the teachings provided herein that any block diagrams presented herein can represent conceptual views of illustrative system components and/or circuitry embodying the principles of the invention. Similarly, one having ordinary skill in the art should appreciate in view of the teachings provided herein that any flow charts, flow diagrams and the like can represent various processes which can be substantially represented in computer readable storage media and so executed by a computer, processor or other device with processing capabilities, whether or not such computer or processor is explicitly shown.

(35) Having described preferred and exemplary embodiments of dual-PPG probes in combination with pulse oximeter monitors and operating methods thereof, (which embodiments are intended to be illustrative and not limiting), it is noted that modifications and variations can be made by

persons having ordinary skill in the art in view of the teachings provided herein, including the appended Figures and claims. It is therefore to be understood that changes can be made in/to the preferred and exemplary embodiments of the present disclosure which are within the scope of the present disclosure and exemplary embodiments disclosed and described herein.

(36) Moreover, it is contemplated that corresponding and/or related systems incorporating and/or implementing the device or such as may be used/implemented in a device in accordance with the present disclosure are also contemplated and considered to be within the scope of the present disclosure. Further, corresponding and/or related method for manufacturing and/or using a device and/or system in accordance with the present disclosure are also contemplated and considered to be within the scope of the present disclosure.

Claims

1. A PPG pulse oximeter, comprising: a dual PPG probe including a circuit configuration of a central PPG sensor connected to a peripheral PPG sensor; and a pulse oximeter monitor connectable to the circuit configuration of the central PPG sensor connected to the peripheral PPG sensor, wherein, when the pulse oximeter monitor is connected to the circuit configuration of the central PPG sensor connected to the peripheral PPG sensor via a connector, the pulse oximeter monitor is configured to: control synchronous generations of a central PPG signal by the central PPG sensor and of a peripheral PPG signal by the peripheral PPG sensor; and control a CPR pulse detection by the dual PPG probe including a detection of a presence of a spontaneous pulse of the central PPG signal and a detection of a presence of a spontaneous pulse of the peripheral PPG signal.
2. The PPG pulse oximeter of claim 1, wherein the central PPG sensor includes a central light emitter optically coupled to a central light detector; wherein the peripheral PPG sensor includes a peripheral light emitter optically coupled to a peripheral light detector; wherein the central light emitter and the peripheral light emitter are connectable to the pulse oximeter monitor in an anti-parallel configuration; and wherein the central light detector and the peripheral light detector are connectable to the pulse oximeter monitor in a parallel configuration.
3. The PPG pulse oximeter of claim 1, wherein the central PPG sensor includes a central light emitter; wherein the peripheral PPG sensor includes a peripheral light emitter; wherein the central light emitter and the peripheral light emitter are connected to the pulse oximeter monitor in an anti-parallel configuration; and wherein the dual PPG probe further includes a common photodetector optically coupled to the central light emitter and the peripheral light emitter.
4. The PPG pulse oximeter of claim 1, wherein the PPG pulse monitor is further configured in connection with the dual PPG probe to: during chest compressions of a CPR, detect the presence of the spontaneous pulse of the central PPG signal and detect the presence of the spontaneous pulse of the peripheral PPG signal.
5. The PPG pulse oximeter of claim 1, wherein the PPG pulse monitor is further configured in connection with the dual PPG probe to: at least one of preceding and succeeding chest compressions of a CPR, detect the presence of the spontaneous pulse of the central PPG signal and detect the presence of the spontaneous pulse of the peripheral PPG signal.
6. The PPG pulse oximeter of claim 1, wherein the PPG pulse monitor is further configured in connection with the dual PPG probe to: measure a differential transit time between a cardiac-induced pulse of the central PPG signal and a cardiac-induced pulse of the peripheral PPG signal.
7. The PPG pulse oximeter of claim 1, wherein the dual PPG probe further includes a sensor identification; and wherein the pulse oximeter monitor is further configured in connection with the dual PPG probe to: identify the connection of the dual PPG probe to the pulse oximeter monitor based on the sensor identification of the dual PPG probe.
8. The PPG pulse oximeter of claim 7, further comprising a pulse oximeter probe including a

sensor identification; and wherein the pulse oximeter monitor is further configured in connection with the pulse oximeter probe to: identify the connection of the pulse oximeter probe to the pulse oximeter monitor based on the sensor identification of the pulse oximeter probe.

9. A PPG pulse monitoring method, comprising: identifying, by a pulse oximeter monitor, a connection of the pulse oximeter monitor to a dual PPG probe including a circuit configuration of a central PPG sensor connected to a peripheral PPG sensor via a connector; controlling, by the pulse oximeter monitor, synchronous generations of a central PPG signal by the central PPG sensor and of the peripheral PPG signal by the peripheral PPG sensor; and controlling, by the pulse oximeter monitor, a CPR pulse detection by the dual PPG probe including detecting a presence of a spontaneous pulse of the central PPG signal and detecting a presence of a spontaneous pulse of the peripheral PPG signal.

10. The PPG pulse monitoring method of claim 9, wherein the central PPG sensor includes a central light emitter optically coupled to a central light detector; wherein the peripheral PPG sensor includes a peripheral light emitter optically coupled to a peripheral light detector; wherein the central light emitter and the peripheral light emitter are connected to the pulse oximeter monitor in an anti-parallel configuration; and wherein the controlling, by the pulse oximeter monitor, synchronous generations of a central PPG signal by the central PPG sensor and of the peripheral PPG signal by the peripheral PPG sensor includes: controlling, by a sensor driver, synchronized activations of the central light emitter and the peripheral light emitter.

11. The PPG pulse monitoring method of claim 9, wherein the central PPG sensor includes a central light emitter; wherein the peripheral PPG sensor includes a peripheral light emitter; wherein the central light emitter and the peripheral light emitter are connected to the pulse oximeter monitor in an anti-parallel configuration; wherein the dual PPG probe further includes a common photodetector optically coupled to the central light emitter and the peripheral light emitter; and wherein the controlling, by the pulse oximeter monitor, synchronous generations of a central PPG signal by the central PPG sensor and of the peripheral PPG signal by the peripheral PPG sensor includes: controlling, by a sensor driver, synchronized activations of the central light emitter and the peripheral light emitter.

12. The PPG pulse monitoring method of claim 9, wherein both the detecting by pulse oximeter monitor of the presence of the spontaneous pulse of the central PPG signal and of the presence of the spontaneous pulse of the peripheral PPG signal are during chest compressions of a CPR.

13. The PPG pulse monitoring method of claim 9, wherein both the detecting by pulse oximeter monitor of the presence of the spontaneous pulse of the central PPG signal and of the presence of the spontaneous pulse of the peripheral PPG signal are at least one of preceding and succeeding chest compressions of a CPR.

14. The PPG pulse monitoring method of claim 9, wherein the controlling, by the pulse oximeter monitor further includes: measuring, by the pulse oximeter monitor, a differential transit time between a cardiac-induced pulse of the central PPG signal and a cardiac-induced pulse of the peripheral PPG signal.

15. The PPG pulse monitoring method of claim 9, further comprising: identifying, by the pulse oximeter monitor, a disconnection of the dual PPG probe from the pulse oximeter monitor and a subsequent connection of a pulse oximeter probe to the pulse oximeter monitor or a disconnection of a pulse oximeter probe from the pulse oximeter monitor and a subsequent connection of the dual PPG probe to the pulse oximeter monitor.
